

AI-Powered Rocket Trajectory Simulation and Maximum Altitude Prediction

A Machine Learning Approach to Aerospace Engineering

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Project Repository: [rocket-trajectory-ml](#)

Submitted as part of the
Open Doors Scholarship Application

Year: 2026

Executive Summary

In this project, I wanted to see if machine learning could predict the maximum altitude of a rocket using data generated from a physics-based simulation. I built a simplified vertical rocket-flight simulator in Python that includes gravity, changing air density with altitude, aerodynamic drag, and changing rocket mass during the flight.

I generated a dataset of 1,000 different rocket configurations by varying six parameters: thrust, initial mass, fuel mass, burn time, drag coefficient, and reference area. I then compared three machine learning models: Linear Regression, Random Forest, and XGBoost.

The best result came from Random Forest after adding physics-based features such as total impulse, thrust-to-weight ratio, mass ratio, and effective drag area. It achieved an R^2 of 0.9749 using 5-fold cross-validation. XGBoost also performed well, with an R^2 of 0.9526 and a Mean Absolute Error of 544 meters.

To check the simulator, I compared its results with an analytical no-drag solution. The difference was only 0.285%, which gave me confidence that the basic numerical simulation was working correctly.

I also analyzed the prediction errors at different altitude ranges. The Mean Absolute Error was 305 m for rockets below 10 km, 801 m for 10–20 km, and 3,324 m above 20 km. The larger error at high altitudes suggests that the model could benefit from more high-altitude examples in the training data.

Finally, I tested the simulator using approximate historical parameters for the German V-2 rocket. The simulation predicted a maximum altitude of 101,109 m, compared with a historical value of 206,000 m. The 50.9% difference is mainly related to the simplified vertical-flight model used in this project, while the real V-2 followed a ballistic trajectory at approximately 45° and involved physical effects that are not included in the current simulation.

Methodology

I divided the project into four main stages: physics simulation, data generation, feature engineering, and machine learning.

1. Physics Simulation

I first built a simplified vertical rocket-flight simulator in Python.

The simulator uses a time step of 0.1 s and models:

- Gravity (9.81 m/s^2)
- Air density changing with altitude using the International Standard Atmosphere model
- Constant thrust during the burn phase
- Aerodynamic drag based on air density, drag coefficient, and velocity
- Fuel consumption and the resulting change in rocket mass

After building the simulator, I checked its results against an analytical no-drag solution. The difference was about 0.285%, which gave me a basic validation of the numerical simulation.

2. Data Generation

I used the simulator to generate 1,000 different rocket configurations.

I varied six parameters:

- Thrust: 30,000–70,000 N
- Initial mass: 800–1,200 kg
- Fuel mass: 250–450 kg
- Burn time: 15–25 s
- Drag coefficient: 0.3–0.7
- Reference area: $0.8\text{--}1.2 \text{ m}^2$

For each configuration, I ran the simulation and recorded the maximum altitude reached.

3. Feature Engineering

After generating the dataset, I added four physics-based features:

- Total impulse = Thrust $\times$ Burn Time
- Thrust-to-weight ratio (TWR)
- Mass ratio
- Effective drag area (CdA)

I added these features to see whether giving the models more physically meaningful information would improve their predictions.

4. Machine Learning

I then trained and compared three regression models:

- Linear Regression as a baseline
- Random Forest
- XGBoost

I evaluated the models using 5-fold cross-validation, with maximum altitude from the physics simulation as the target variable.

This allowed me to compare the models using the same dataset and also test whether the physics-based features improved their performance.

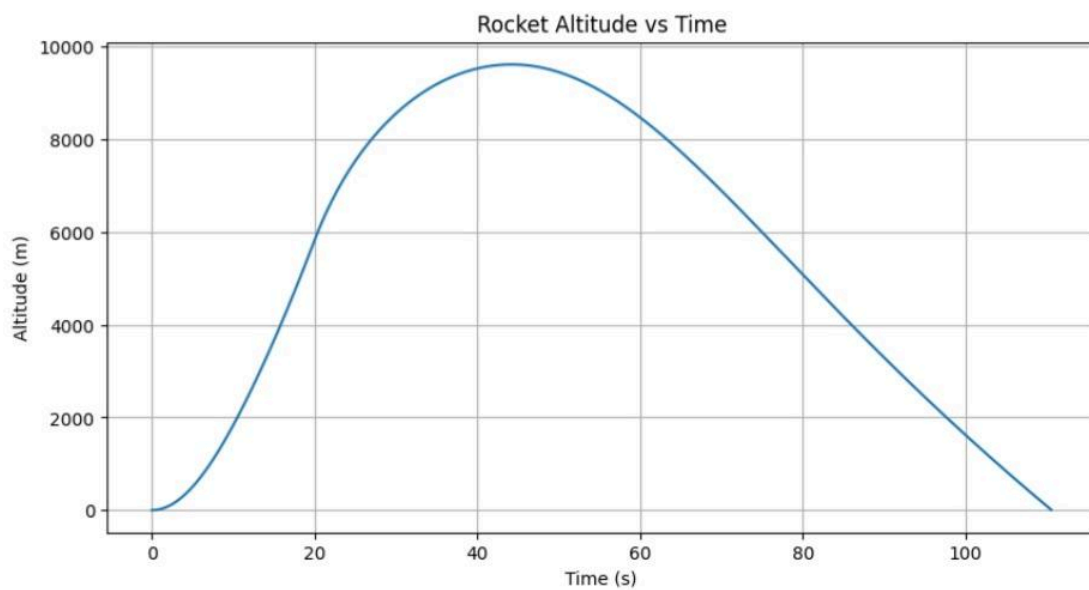


Figure 1: Simulated altitude of a test rocket over time.

Model Evaluation

I compared three models using 5-fold cross-validation with the 1,000 simulated rocket configurations.

The results were:

Model comparison (R^2 scores):

Model	Raw Features (R^2)	Physics Features (R^2)
Linear Regression	0.9015	0.9302
Random Forest	0.9451	0.9749
XGBoost	—	0.9526

- Random Forest gave the best result, with an R^2 of 0.9749 after adding the physics-based features. XGBoost also performed well, with an R^2 of 0.9526 and a Mean Absolute Error of 544 meters.

I also analyzed the feature importance of the Random Forest model to see which parameters had the greatest influence on the predicted maximum altitude. This helped me understand which rocket parameters were most important for the model's predictions.

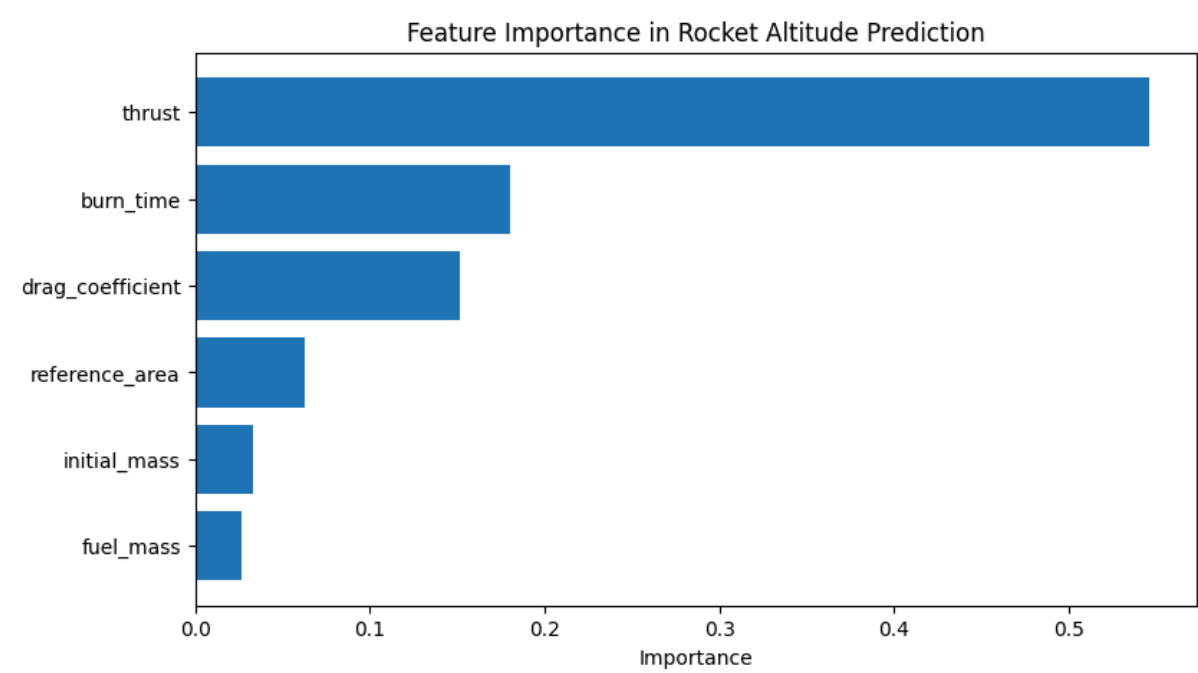


Figure 2: Feature importance in predicting maximum altitude.

Error Analysis

I analyzed the prediction errors of the XGBoost model across different altitude ranges.

Altitude Range	Mean Absolute Error
Low (<10 km)	305 m
Mid (10–20 km)	801 m
High (>20 km)	3,324 m

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The model was more accurate at lower altitudes and less accurate at higher altitudes.

The higher error at high altitudes may be related to the smaller number of high-altitude examples in the generated dataset. This suggests that adding more high-altitude rocket configurations could help improve the model's performance in this range.

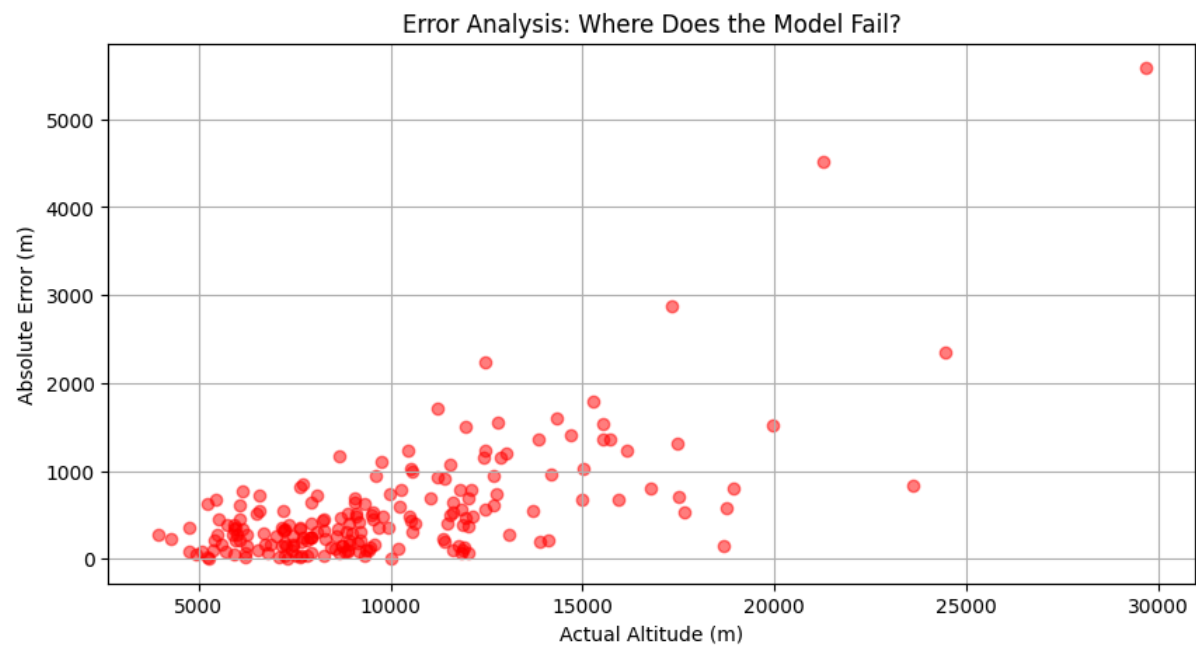


Figure 3: Prediction error as a function of actual altitude.

V-2 Case Study

To test my simulator on a real-world example, I used approximate historical parameters for the German V-2 rocket.

Parameter	Value
Simulated maximum altitude	101.109 m
Historical maximum altitude	206,000 m
Relative error	50.9%

The difference is quite large, but this was expected because my simulator assumes vertical flight only. The real V-2 followed a ballistic trajectory at approximately 45° and was affected by physical factors that are not included in my simplified model.

This comparison showed me one of the main limitations of the current simulator and also gave me a clear direction for improving it in future versions.

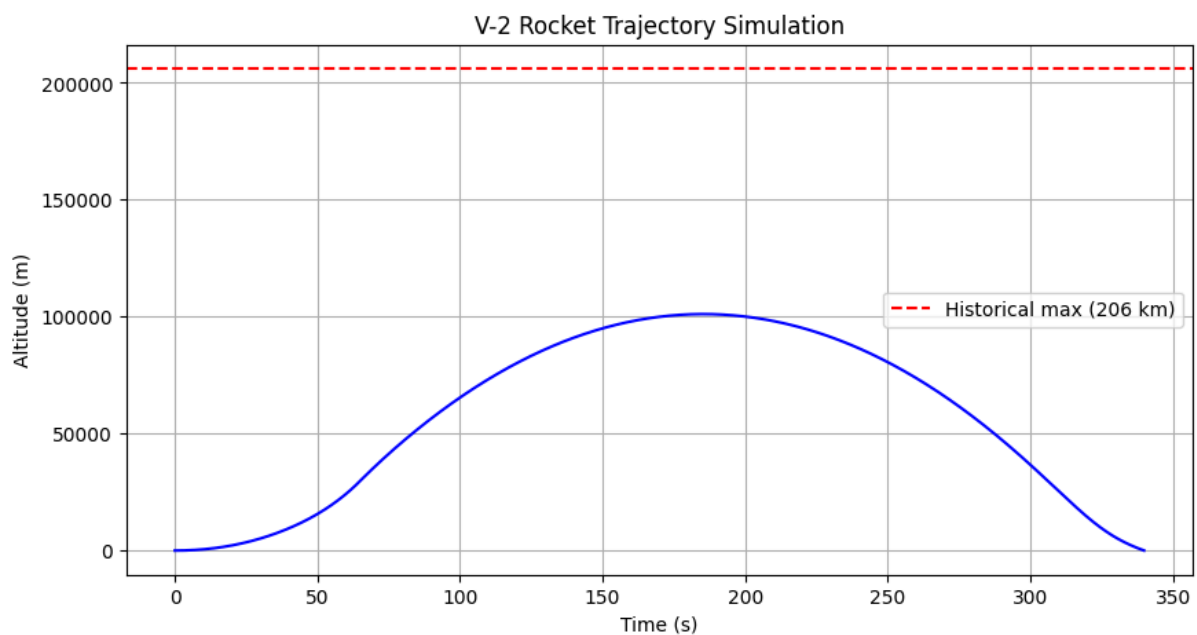


Figure 4: Simulated V-2 trajectory vs. historical altitude.

Limitations

This project is a simplified educational simulation rather than a high-fidelity rocket-flight model.

The main limitations of my current model are:

- Vertical flight only, with no launch angle
- Simplified atmospheric model, without wind or weather effects
- Constant gravity
- Constant drag coefficient, without Mach-number dependency
- No Coriolis effect
- No multi-stage rockets
- Synthetic training data generated by the simulator itself
- No real flight data

Because of these limitations, the results should be considered as a learning and experimentation tool rather than an accurate representation of real rocket flight.

Future Work

There are several improvements I would like to explore in future versions of this project:

- Extend the simulator to 2D or 3D trajectories and include launch-angle optimization
- Use a more realistic atmospheric model
- Add Mach-dependent drag
- Integrate real rocket-flight data for validation
- Explore Physics-Informed Neural Networks (PINNs)
- Investigate Reinforcement Learning for trajectory optimization and correction
- Compare additional ensemble and machine-learning methods

These improvements could make the simulation more realistic and allow me to explore more advanced applications of machine learning in rocket trajectory prediction and optimization.

Conclusion

This project showed me that machine learning can be used to predict a rocket's maximum altitude from data generated by a physics-based simulation. The best result was achieved with Random Forest and physics-based features, reaching an R^2 of 0.9749.

In this project, I combined classical mechanics, numerical simulation, feature engineering, and machine learning in one workflow. I also learned how important it is to validate simulation results, check prediction errors, and understand the limitations of a model before interpreting its result

Beyond the technical results, this project strengthened my interest in aerospace engineering and machine learning. I would like to continue developing the project by making the simulation more realistic, using real flight data, and exploring more advanced machine-learning methods in future work.